INTRODUCTION

Statistics for Computer Engineering

Lecture time

- Tuesday 13.30-15.30
- Thursday 10.00-12.00 & 13.30-15.30
- TAs will be around to help any questions
  - Tuesday 15.30-16.00
  - Thursday 9.00-10.00 & 15.30-16.00

Communications

Mycourseville

2110205
(2024/1)
https://www.mycourseville.com?q=courseville/course/register/2110205_2024_1

password: uma

Discord

https://discord.gg/kkTNHxqM

Format

On-site

Summary video for review

Homework assignments

In class activity/quiz

Midterm/final

Quiz and Exams

- การสอบทั้งหมด in-person

Homework

- Out: Thursday
- Due: Wednesday night
- Only a couple of problems in the homework will be graded (will be declared for each homework)
- The rest should be used for practice
- Solutions will be provided after 24 hours of the deadline
- Submission
  - One single PDF only. Can write or type, but final submission must be in PDF format. (No zip, jpg, png, etc.). If it's a coding homework, print out in PDF format.

Getting help

- Discord
  - Text Channels/Question channel
    - Direct PMs will be ignored if not a personal question
- Off hours discord
  - Schedule TBA
- No help will be offered on Wednesdays
  - Do your homework early!

Academic integrity

- You shall not copy others.
  - For homeworks, violation will result in **0** for both parties.
  - For quizzes, violations result in an **F** and other penalties according to Faculty of Engineering guidelines.
- You can collaborate or discuss but:
  - You must declare the person/Chat-GPT you collaborate with in the homework
  - You shall not show exact write-up with each other
    - Can discuss ideas but not implementation
    - Can compare final answers

Learning theory

- Education research have shown that
  - Mental effort and active learning benefits student learning.

ChatGPT

- Access to ChatGPT can harm learning outcome.

Generative AI Can Harm Learning

59 Pages • Posted: 18 Jul 2024

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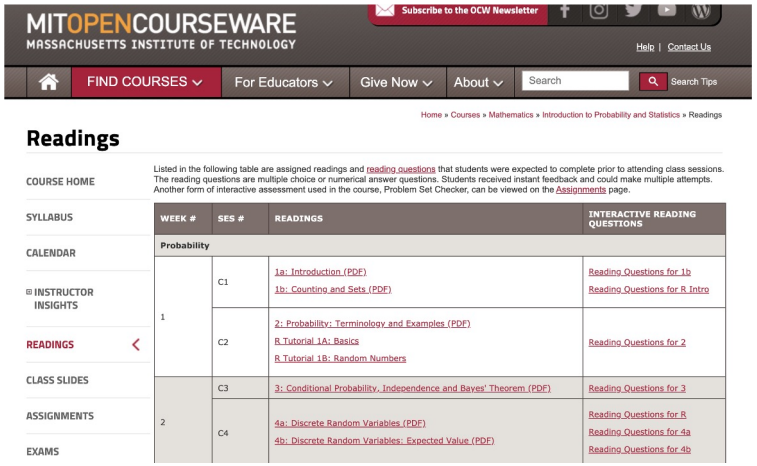
Date Written: July 15, 2024

Responsible use of ChatGPT

- Human goes first
  - Think about the problem and try to solve it your own.
  - After solving, have ChatGPT critique your ideas/solutions

Books



https://ocw.mit.edu/courses/mathematics/18-05-introduction-to-probability-and-statistics-spring-2014/readings/

INTRODUCTION

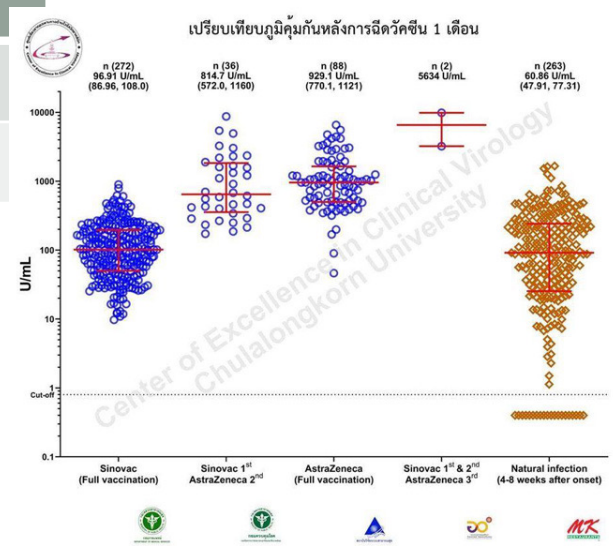
Statistics for Computer Engineering

Motivation

- Many questions we want to answer that is related to probability
  - COVID drug, should it be used?

	Dead	Alive	Total
Drug A	9	31	40
No drug	10	30	40

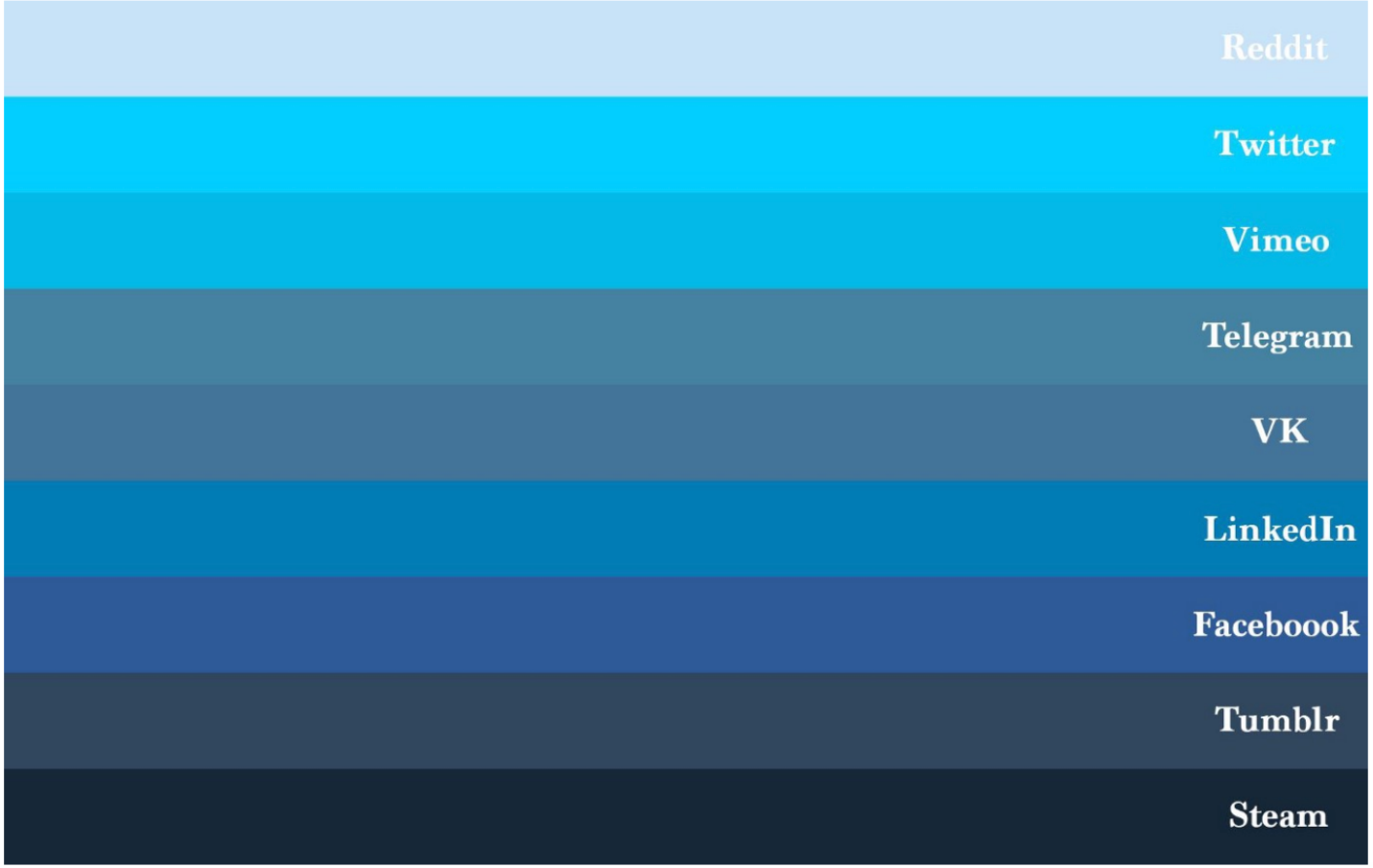
	Dead	Alive	Total
Drug A	0	5	5
No drug	10	30	40



Data Driven Decisions: Google's '50 shades of Blue' experiment

Would you believe it if I told you that switching to a different shade of blue netted Google as additional $200 Mn in Ad revenue?

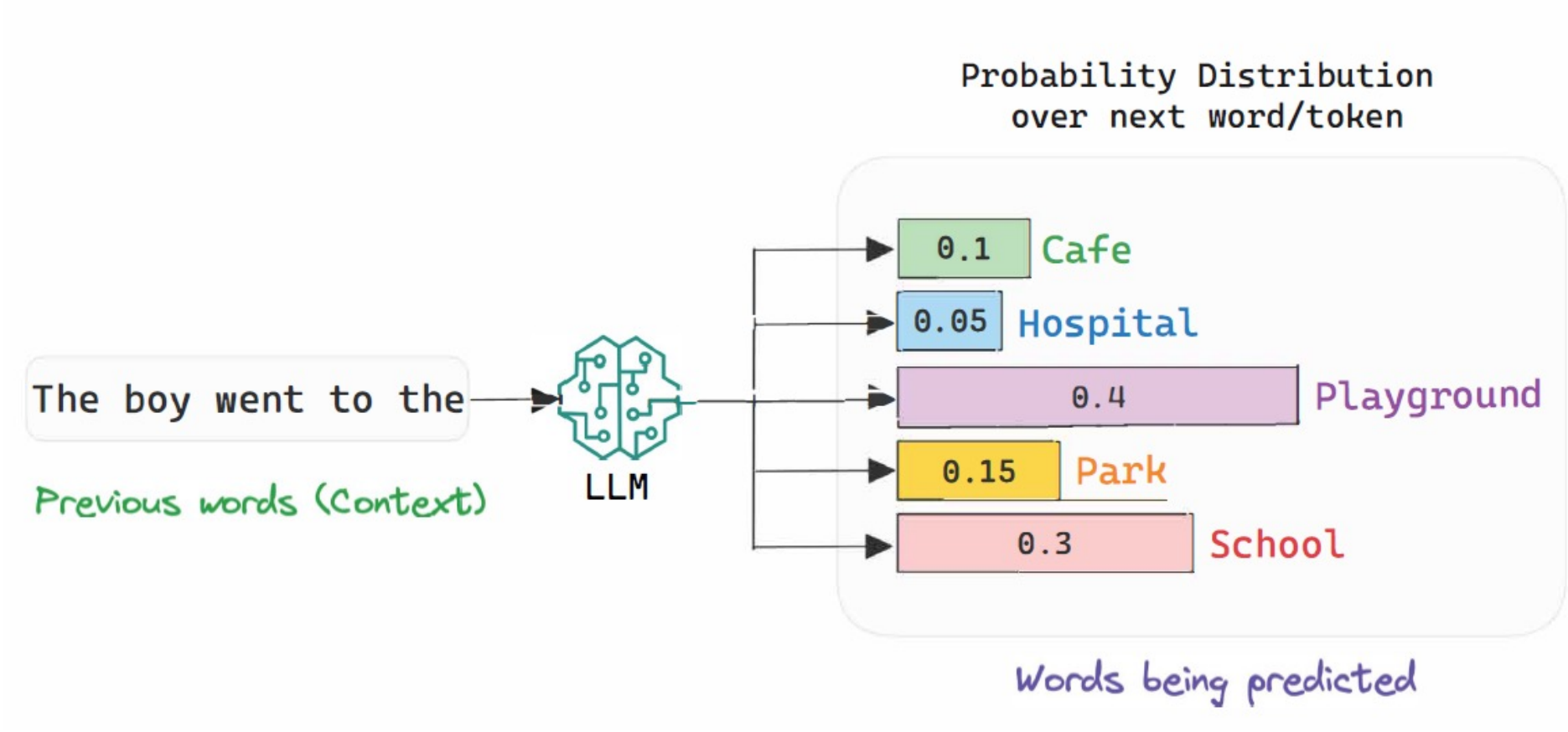
Bharath Kumar Balasubramanian Mar 16 · 3 min read



https://medium.com/@BBharathKumar/data-driven-decisions-googles-50-shades-of-blue-experiment-996f01819a97

The shades of blue across other social platforms. https://imgur.com/gallery/uzzls

Machine learning



https://www.linkedin.com/pulse/how-do-language-modelsllm-work-we-call-chatgpt-mishra-fdqsc/

Insurance pricing

- Life insurance payout = 1000000
- Probability of living one more year = 0.995

- What's the break even price for the insurance?

Probability vs Statistics I

- Different subjects: both about random processes

- Probability

Toss a fair coin. What's the probability of getting 2 heads in 3 tosses?

- Statistics

You toss a coin of unknown bias 100 times. You got 60 heads. Draw some conclusion from this data.

Known process vs Infering from the outcome

Probability vs Statistics II

- Probability
  Toss a fair coin. What's the probability of getting 2 heads in 3 tosses?

Rules and logics concerning probability calculation. One correct answer.

- Statistics
  You toss a coin of unknown bias 100 times. You got 60 heads. Draw some conclusion from this data.

More like an art. Make decisions based on possibly different assumptions. No single correct answer.

What is Probability?

- Frequentist

  Probability = rate of occurrence in a large number of trials

  High probability means high rate of occurrence

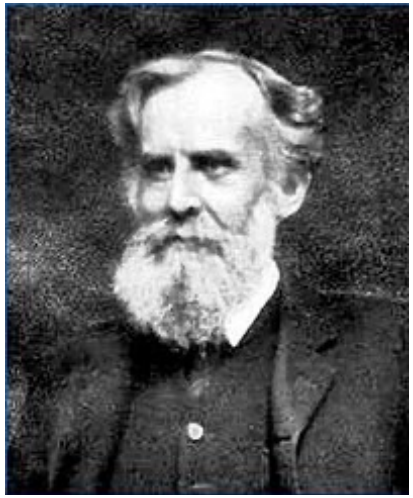
  Often used in scientific fields

- Bayesian

  Probability = uncertainty in your knowledge of the world

  High probability means high degree of belief

  Often used in modern analysis, and machine learning



Venn

Bayes

Bayesian vs Frequentist

- What is the probability of the being head?

Summary

- Probability vs Statistics
- Frequentist vs Bayesian

- Readings MIT 1a

- Next time
  - Python, numpy, pandas, scipy, matplotlib